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MILITARY SPECIFICATION

COMPENSATOR, INERTIA, FOR AS-473/ASB-1
(BuOrd Drawing 790394 Inertia Compensating Unit)

1. SCOPE

1.1 This specification covers the equipment, material, and performance requirements to be used in the manufacture of an Inertia Compensator for use on a radar scanning antenna. One type and one grade only is specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, DRAWINGS, AND PUBLICATIONS

2.1 The following specifications, standards, drawings, and publications of the issue in effect on date of invitation for bids form a part of this specification.

2.1.1 Specifications

Military

MIL-P-116	Preservation, Methods of

2.1.2 Standards

Military

MIL-STD-129	Military Standard Marking of Shipments

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2.1.3 Drawings

Bureau of Ordnance, Department of the Navy

List of Drawings, LD 272216, Inertia Compensating Unit and all drawings and specifications listed thereon.

2.1.4 Publications

Bureau of Ordnance, Department of the Navy

NAVORD OP 400	Manufacture and Inspection of Ordnance Material, General Specification for
NAVORD OD 8176	Paragraphs applicable to this equipment

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the Procuring Agency or as directed by the Contracting Officer)

2.1.5 Other Publications

National Bureau of Standards Publications

Handbook H28	Screw-Thread Standards for Federal Services

(Copies of H28 may be obtained upon application, accompanied by money order, coupon, or cash to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

3. REQUIREMENTS

3.1 Type Tests - Type tests as specified in Section 4 of this specification shall be made on representative samples of production assemblies.

3.2 General - Equipment furnished under this specification shall meet applicable requirements listed in NAVORD OD 8176. In cases of conflict, this specification shall apply.

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3.3 Material - Materials shall conform to applicable specifications as specified herein and on the drawings. When materials are used which are not specifically designated, they shall be of the best commercial quality and entirely suitable for the purpose.

3.4 Design and Construction - The assembly shall conform to and be approved in accordance with this specification and applicable specifications, standards, drawings, and publications.

3.5 Interchangeability - Interchangeability shall be insured by conformance with dimensional limitations set forth on outline drawing, BuOrd number 1332413.

3.6 Nameplates - A nameplate, permanently and legibly marked and made in accordance with BuOrd drawing 776416, shall be attached to each assembly.

3.7 Performance

3.7.1 The center tooth of the internal sector, BuOrd drawing 759914, shall engage the center tooth socket of the double sector gear, BuOrd drawing 759915. See point "D" of Figure 1.

3.7.1.1 Backlash between internal sector and the double sector gear shall not exceed 0.002 inch. See point "D" of Figure 1.

3.7.2 The maximum angular displacement of the double sector, BuOrd drawing 759915, shall be 30° 10' ± 0° 17' either side of the centerline, as shown in Figure 1.

3.7.2.1 With the double sector at its maximum angular displacement, each rotor shaft shall be parallel to the mounting surface "C" within 0° 5'.

3.7.2.2 When the double sector is positioned as in 3.7.2, the angle "M" of the cam, Figure 1, shall be 5° 0' ± 0° 30'.

3.7.3 When the center tooth "L" of the double sector is at its maximum angular displacement to the left of the centerline as shown in Figure 1, gear "F_1" shall be at the top of the figure as shown and gear "F_2" shall be at the bottom of the figure as shown.

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3.7.4 When the center tooth "L" of the double sector, Figure 1, is centered over shaft "J" within 0.001 inch, the internal pointer shall be centered on target "N" within 0.010 inch.

3.7.5 At points "F_1" and "F_2" of Figure 1, the backlash shall be 0.0005 inch minimum to 0.002 inch maximum with 3 pounds end load applied to cardans in direction "Z_1" and "Z_2" respectively.

3.7.6 At points "G_1" and "G_2," Figure 1, backlash shall be 0.0005 inch minimum to 0.002 inch maximum with 3 pounds end load applied to cardans in direction "Z_1" and "Z_2" respectively.

3.7.7 At point "J," Figure 1, end play shall be 0.002 inch maximum under 2 pounds reversing axial load.

3.7.8 At points "K_1" and "K_2." Figure 1, axial play shall not exceed 0.003 inch maximum under 5 pounds reversing axial load.

3.7.9 End play of rotor shafts shall not exceed 0.003 inch under 2 pounds reversing axial load.

3.8 Environmental Conditions

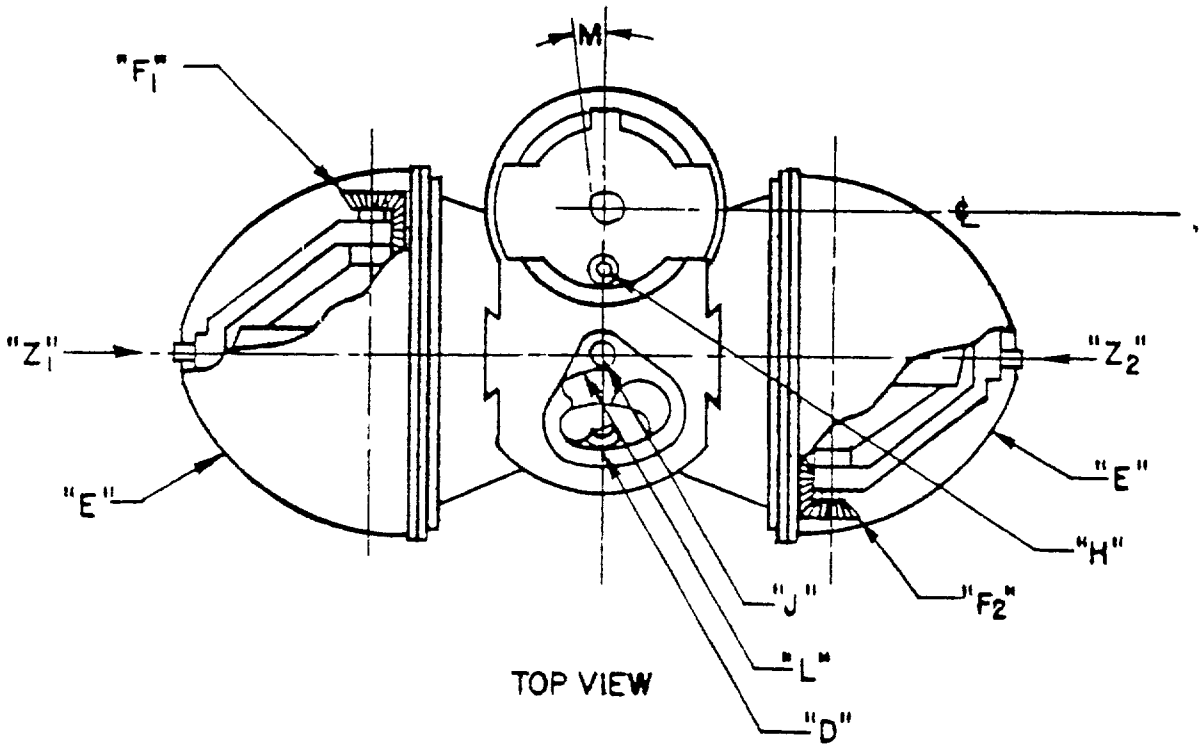
3.8.1 Temperature - The equipment shall conform with all requirements of this specification under temperature conditions as specified in NAVORD OD 8176 except that the temperature range shall be modified to read -55°C to +71°C (-67°F to +160°F).

3.8.2 Vibration - The equipment shall be subjected to vibration conditions as specified in OD 8176 except that the total excursion shall not exceed 0.031 inch. See BuOrd drawing 1332413 for normal mounting.

3.8.3 Shock - The equipment shall not suffer damage or subsequently fail to provide the performance specified herein when it is mounted as normally used and is subjected to six (6) impact shocks of 10g's, each shock impulse having a time duration of 11 ± 1 milliseconds. See BuOrd drawing 1332413 for normal mounting. Two (2) shocks shall be applied in each of the three (3) principal axes, one (1) shock in each direction.

3.9 Balance - The unit shall be statically balanced in accordance with drawing requirements.

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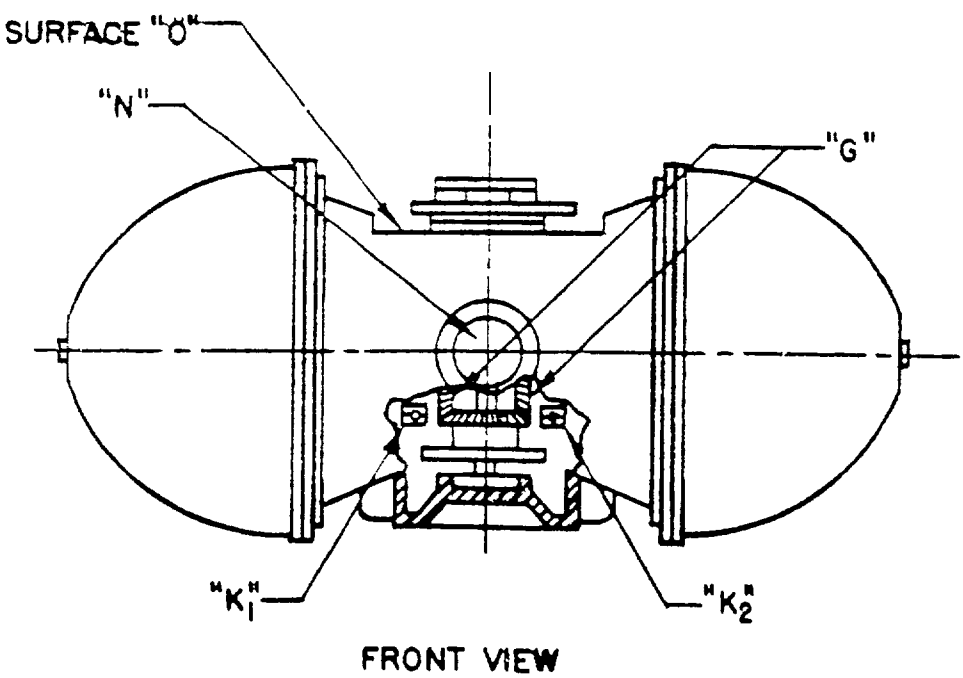


Figure 1. Reference Points

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3.10 Lubrication - All gearing shall be lubricated with instrument grease per MIL-G-15793 (BuOrd).

3.11 Workmanship - Workmanship shall be in accordance with the best instrument practice. Particular attention shall be given to machined surfaces to insure conformance with drawing limitations and against superficial defects. Attention shall also be given to painted areas to insure clean and scratch-free surfaces.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 General Instructions

4.1.1 Responsibility for Tests - The Contractor shall furnish all samples, if required, and shall be responsible for accomplishing the inspections and tests specified herein.

4.1.2 Previous Approval - Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.1.3 Supervision of Tests - When inspection is conducted at the Contractor's plant, all inspection and testing shall be under the supervision of a Government Inspector.

4.1.4 Record of Tests - The Contractor shall maintain a record available to the Inspector, showing the results of all Inspection Tests.

4.1.5 Laboratory Facilities - Contractors not having laboratory facilities satisfactory to the Inspector shall engage the services of a commercial testing laboratory acceptable to the Inspector.

4.1.6 Pre-Submission Testing - No item, part, or complete equipment shall be submitted to the Government Inspector by the Contractor until it has been previously tested and inspected by the Contractor and found to comply to the best of his knowledge and belief with all applicable requirements. With the consent of the Contractor and at the discretion of the Procuring Agency, this prior test and inspection may be participated in or witnessed by the Government Inspector with the object of eliminating the necessity of repeating such test and inspection after the equipment has been formally submitted to the Government Inspector.

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4.1.7 Rejection and Retests - The equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished tne Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Procuring Agency.

4.2 Inspection

4.2.1 Examination of Product - Each assembly, subassembly, and part shall be examined as the Inspector may deem necessary to determine conformance witn this specification and tne applicable drawings with respect to material, dimensions, instructions, and workmanship.

4.3 Test Procedures

4.3.1 Classification - Tests shall be classified as follows:

a. Type Tests
b. Inspection Tests

4.3.2 Type Tests - Type tests shall be performed on one (1) of tne first twenty (20) production units and on one (1) of each subsequent group of one hundred (100) production units or parts thereof, for each manufacturer supplying equipment in accordance with this specification.

4.3.2.1 Type test samples shall be selected at random from production units tnat have passed all inspection tests and shall nave inspection tests repeated after being subjected to environmental conditions specified in 3.8 of tnis specification.

4.3.3 Inspection Tests - Inspection tests shall be made on all production assemblies submitted for acceptance under the Contract, to determine conformance with the requirements of tnis specificatior and to determine tnat tne assemblies are equivalent in performance and construction to the approved samples. All production lots as defined in 4.3.2 will be accepted contingent upon type tests approval.

4.3.3.1 Test Equipment - The following listed equipment or their equivalents will be required to perform tests specified in 4.3.3.2.

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4.3.3.1.1 Test fixture on which to mount the Inertia Compensator while making measurements.

4.3.3.1.2 Two (2) fixtures, one to support each cardan in the same manner as the end caps. The fixtures shall be of open construction so measurements may be taken. These fixtures shall also provide a means of applying 3 pounds axial end load to the cardans.

4.3.3.2 Tests - Conduct tests on all assemblies at normal room conditions to determine conformance with 3.7.1 through 3.7.9.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Packing - Packaging and packing shall be in accordance with MIL-P-116 , Method II, paragraph 3.4.7.2.2 or 3.5.7.2.4, as applicable. Such items shall be packaged by either of the above methods as directed by the Procuring Agency. Further, metal containers shall be surrounded by suitable cushioning material and packed in a wooden case as directed by the Procuring Agency.

5.1.1 Labeling and Marking - Labeling and marking shall be in accordance with MIL-STD-129, Military Standard Marking of Shipments, except as may be modified by the Procuring Agency.

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